Outliers

Causes

Human errors — Data entry errors

Instrument errors — Measurement errors

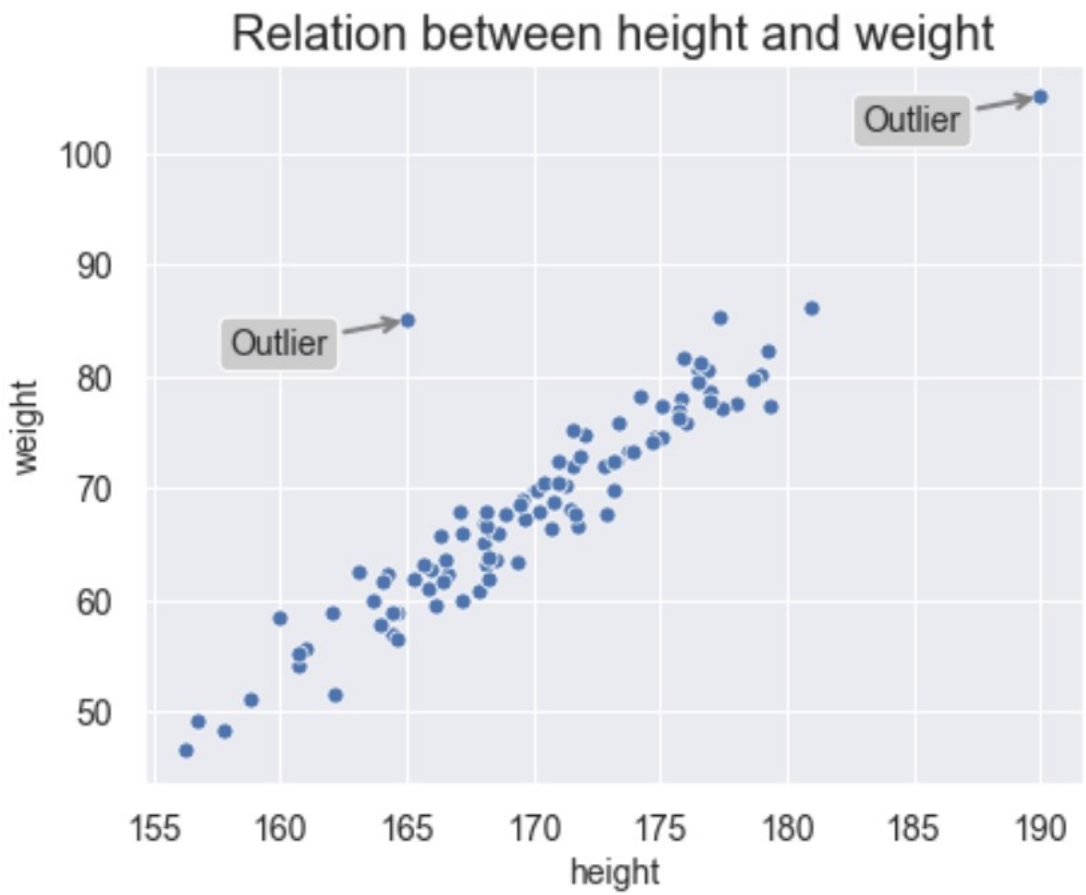
Experimental errors — data extraction or experiment planning/executing errors

Intentional — dummy outliers made to test detection methods

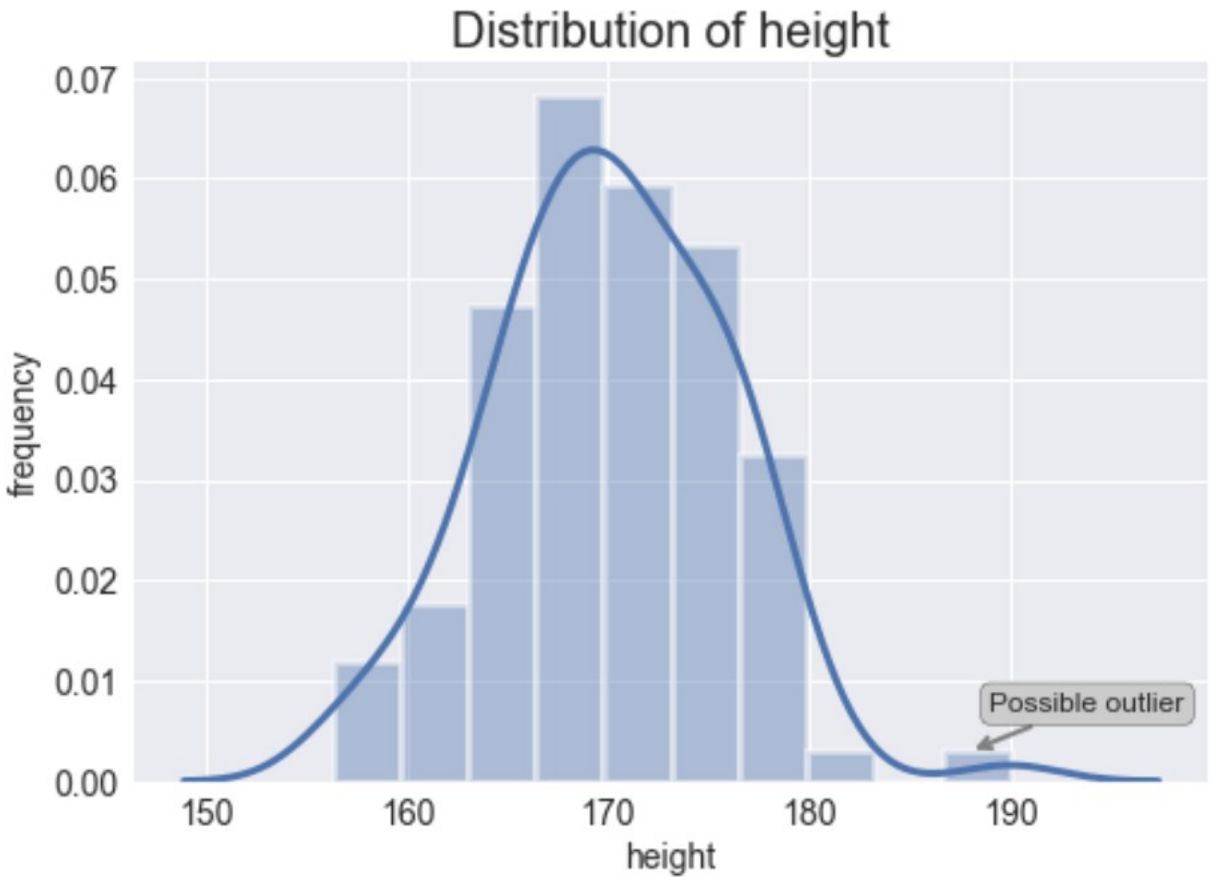
Data processing errors — data manipulation or data set unintended mutations

Sampling errors — extracting or mixing data from wrong or various sources

Natural — not an error, novelties in data



Histogram


Distribution of height
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0.00
frequency
150
160
170
180
190
height
Possible outlier

Box Plots

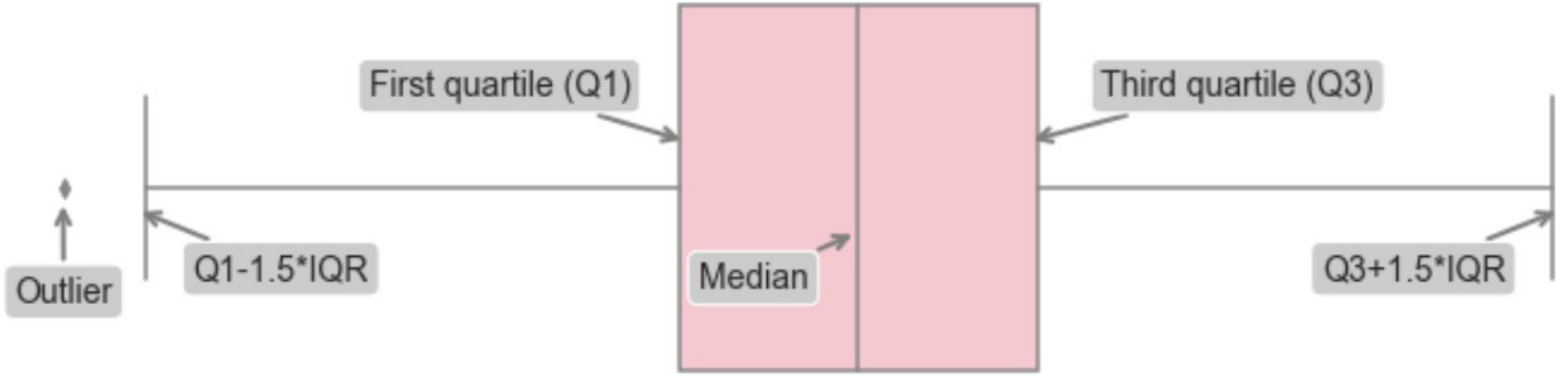
The minimum: the lowest value excluding outliers (calculated according to the IQR proximity rule)

The maximum: the largest value excluding outliers (calculated according to the IQR proximity rule)

The median (Q2): the midpoint of the distribution

The first quartile (Q1): the midpoint of the lower half of the data set

The third quartile (Q3): the midpoint of the upper half of the data set

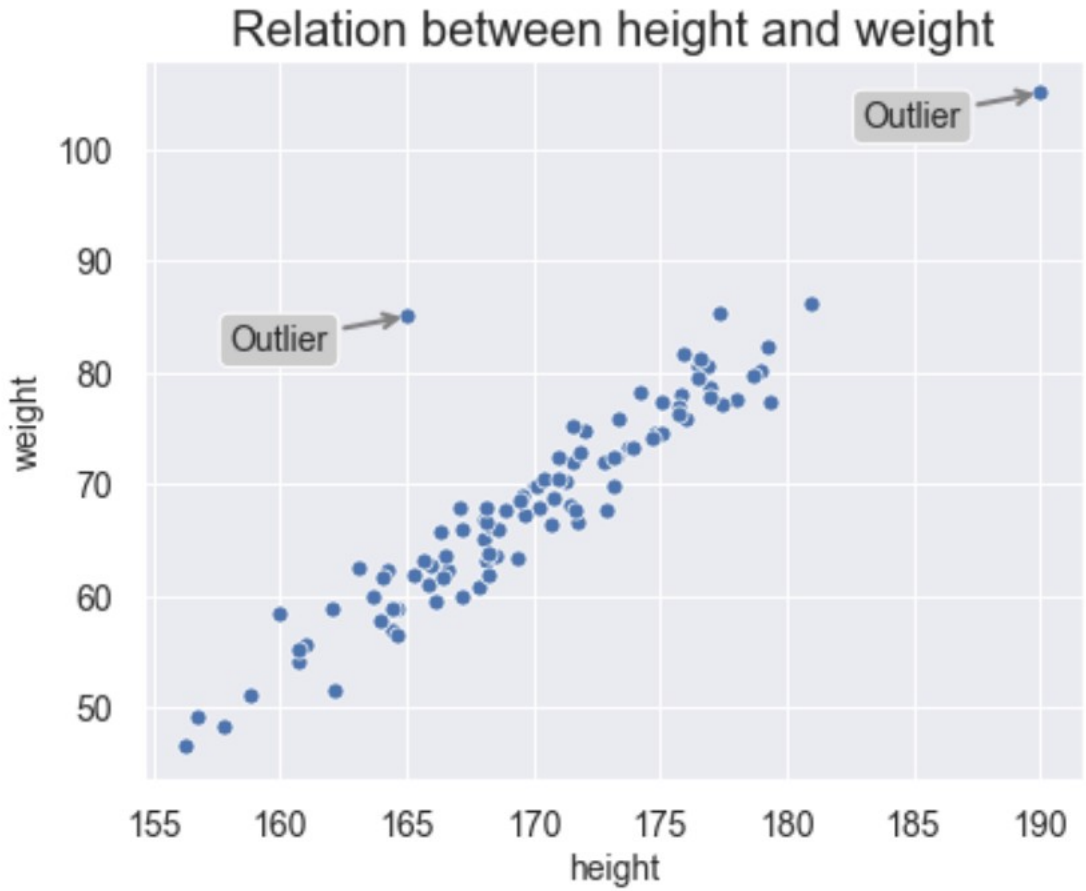


IQR = Q3 – Q1

Upper boundary = Third quartile + (1.5 * IQR)

Lower boundary = First quartile — (1.5 * IQR)

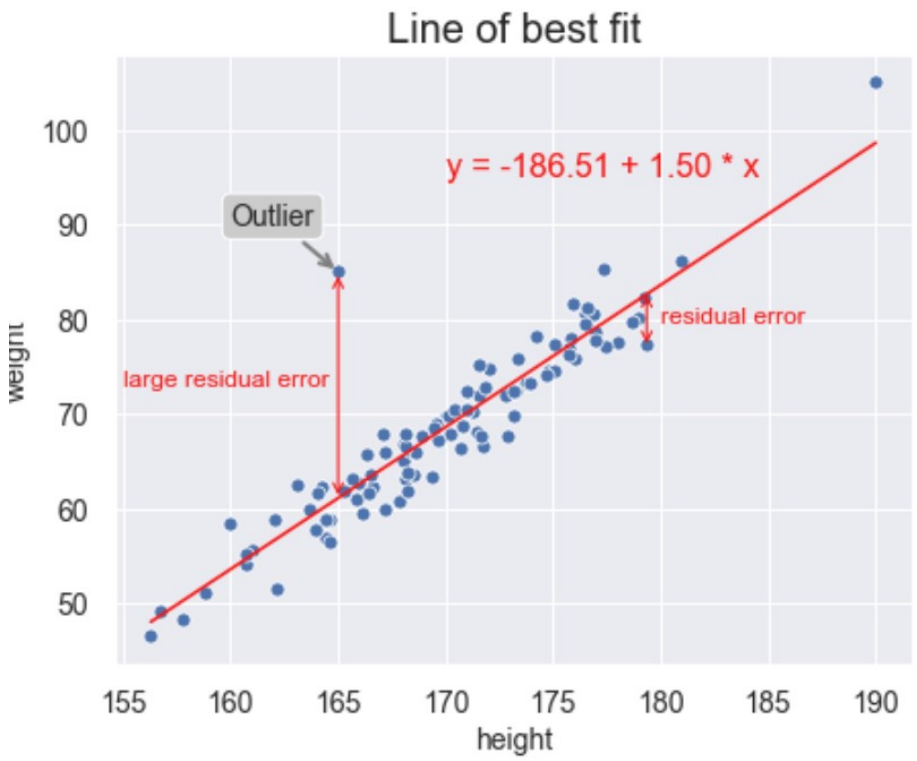
Scatter Plot

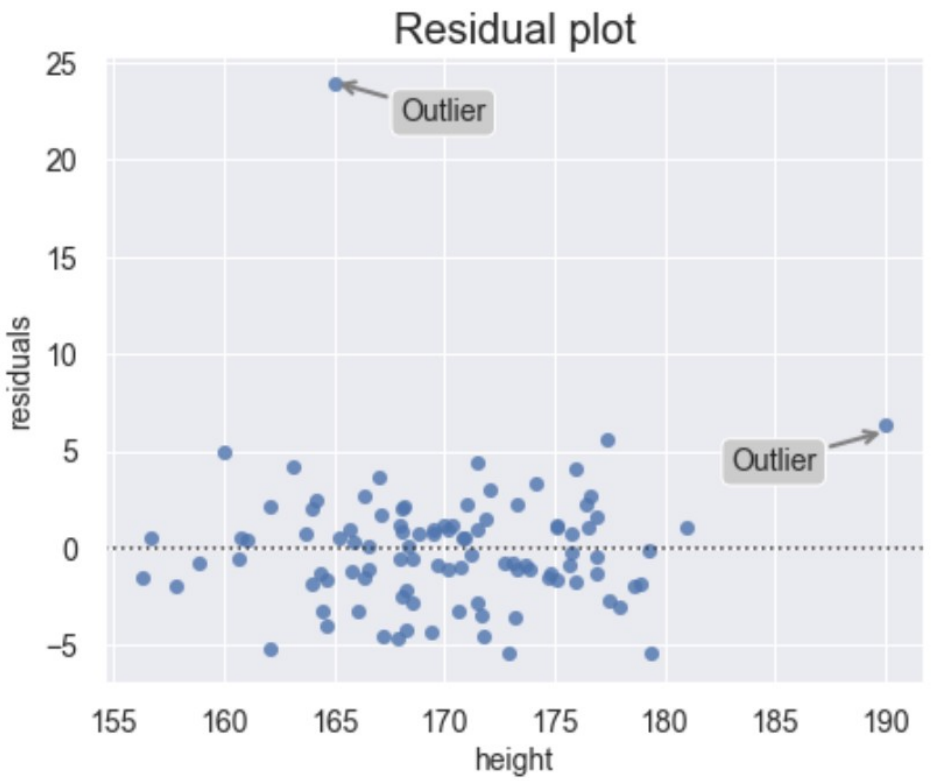


Residuals

Simple linear regression:
y = B0 + B1 * x

residual error = y — y_hat





Applications

https://towardsdatascience.com/outlier-detection-theory-visualizations-and-code-a4fd39de540c

- **Fraud detection** — detecting fraudulent applications for credit cards, state benefits or detecting fraudulent usage of credit cards or mobile phones.
- **Loan application processing** — to detect fraudulent applications or potentially problematic customers.
- **Intrusion detection** — detecting unauthorized access in computer networks.
- **Activity monitoring** — detecting mobile phone fraud by monitoring phone activity or suspicious trades in the equity markets.
- **Network performance** — monitoring the performance of computer networks, for example, to detect network bottlenecks.
- **Fault diagnosis** — monitoring processes to detect faults in motors, generators, pipelines, or space instruments on space shuttles for example.
- **Structural defect detection** — monitoring manufacturing lines to detect faulty production runs for example cracked beams.
- **Satellite image analysis** — identifying novel features or misclassified features.
- **Detecting novelties in images** — for robot or surveillance systems.
- **Motion segmentation** — detecting image features moving independently of the background.
- **Time-series monitoring** — monitoring safety-critical applications such as drilling or high-speed milling.
- **Medical condition monitoring** — such as heart-rate monitors.
- **Pharmaceutical research** — identifying novel molecular structures.
- **Detecting novelty in the text** — to detect the onset of news stories, for topic detection and tracking or for traders to pinpoint equity, commodities, FX trading stories, outperforming or underperforming commodities.
- **Detecting unexpected entries** in databases — for data mining to detect errors, frauds, or valid but unexpected entries.
- **Detecting mislabelled data** in a training data set.